

BEARING WITNESS.



A REAL BEARING (SKF 129 HA) · PHOTO: WIKIMEDIA COMMONS, CC BY 4.0

41%

of all motor failures are bearings. The largest utility motor reliability study ever run: 6,312 motors.

ALBRECHT ET AL., IEEE TRANS. ENERGY CONVERSION, 1986

And most plants still check them once a month, by hand .

\$125,000

median cost of ONE HOUR of unplanned downtime. Two-thirds of plants eat at least one a month.

ABB, VALUE OF RELIABILITY SURVEY, 2023 · N = 3,215 PLANTS

A bearing can die inside the gap between two readings. The fix everyone sells is cloud, and this data is not allowed to leave the building .

AN ALWAYS-ON SCREENING AGENT THAT LIVES ENTIRELY ON THIS BOX.

It watches every window of vibration, explains what it sees with evidence a human can check, and files the work order.

Nothing leaves the room.

PHYSICS DOES THE DIAGNOSIS. A LOCAL MODEL FILES THE PAPERWORK. A HUMAN SAYS YES.

01 · DETECT

Persistent change against the asset's own early baseline.

02 · EXPLAIN

Known machine frequencies labelled before anything is called a fault.

03 · CORROBORATE

Two independent spectral views must agree, at slip-aware widths.

04 · ESCALATE

A typed work order with evidence locators. Approve, reject, or defer.

BUSINESS VALUE • SOMETHING A COMPANY WOULD ACTUALLY PAY FOR

ONE CATCH INSIDE THE MONTHLY BLIND SPOT **PAYS** **FOR THE BOX.**

THE BUYER

Every OT-isolated plant that already refuses cloud monitoring: pharma, food, defense, chemicals, utilities. The customer who needs this most is exactly the one nothing else can serve.

THE PLANNER'S MORNING

Before: a monthly route, a clipboard, a guess.

After: a screening queue with evidence, and the work order already in the system of record.

AT ABB'S MEDIAN DOWNTIME RATE OF \$125,000/HOUR (2023 SURVEY, N=3,215)

THE SECURITY TEAM IS THE REAL BUYER, AND WE ARRIVE WITH THE AUDIT READY.

A CLOUD VENDOR

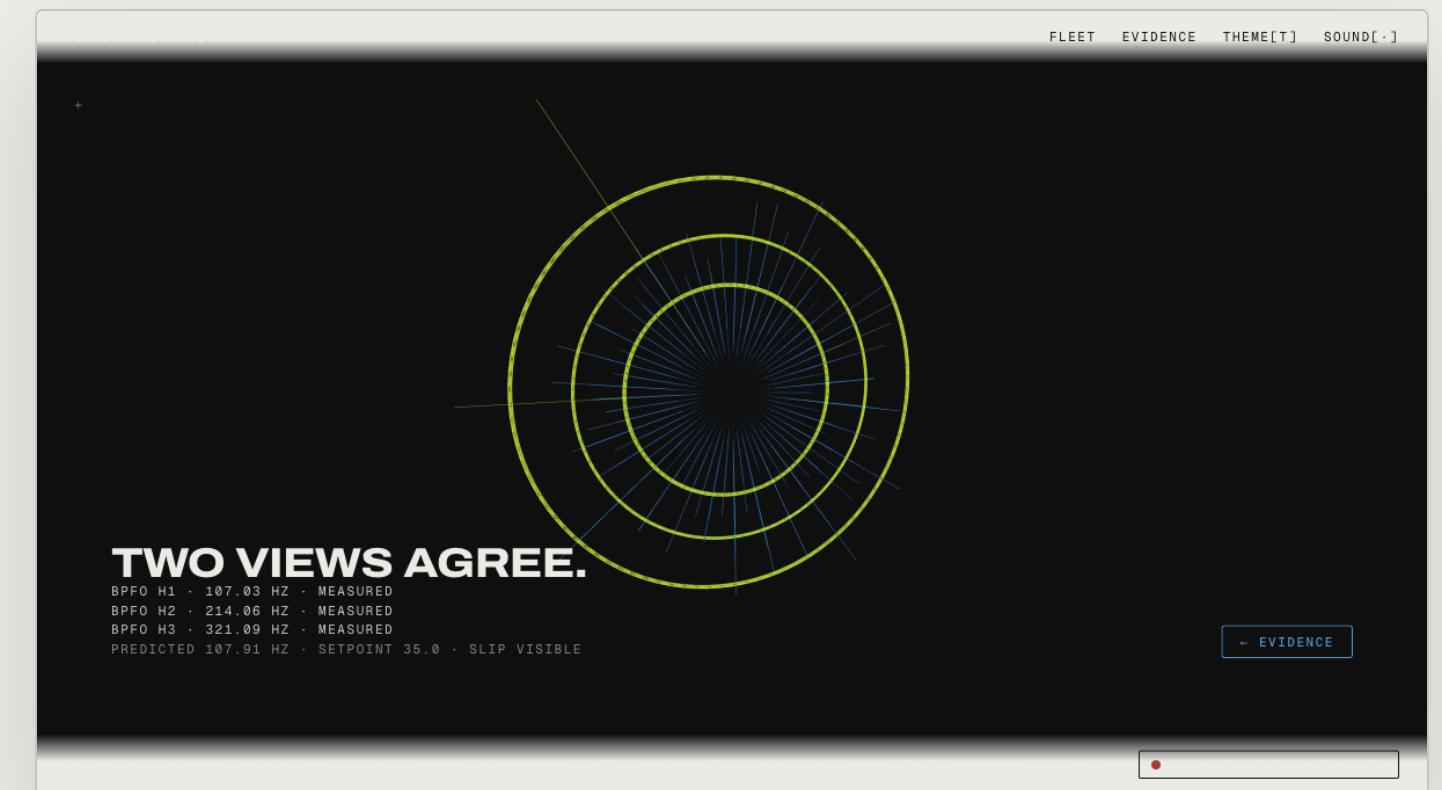
Months in vendor security review. The data leaves the building, so for an OT-isolated plant the answer is no before the demo starts. A subscription that never ends.

THIS BOX

The review is checkable in an afternoon: the policy lists zero cloud hosts, and the agent works with cloud routes firewalled, live. One box, bought once. One catch pays for it.

TECHNICAL EXECUTION · IT DOESN'T BREAK

- NVIDIA NemoClaw + OpenClaw + OpenShell (deny-by-default egress, shown live)
- Local Qwen3 27B (4-bit, Ollama) on the GB10 · MongoDB Community, localhost only
- SciPy owns diagnosis: deterministic, auditable, frozen thresholds
- 150+ tests · CI green · runs from a cold clone



THE CORROBORATION VIEW · WEBGL FED BY THE REAL ENVELOPE BINS

**DETECT THE CHANGE.
EXPLAIN THE SPECTRUM.
ESCALATE WITH EVIDENCE.
A HUMAN HAS TO SAY YES.**

BUILT TODAY, ON THE BOX, ON REAL RUN-TO-FAILURE DATA · IT NEVER GUESSED ONCE

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[GITHUB.COM/STEPHENSOOK/BEARING-WITNESS](https://github.com/stephensook/bearing-witness) · MIT